

# Shell Scripting, Automation & Web Servers

## Shell Scripting

- Shell Scripting is the process of writing a sequence of Linux commands in a file to automate tasks, it's simple referred to as Script.
- A shell script is executed by a shell such as Bash and helps reduce manual effort while improving efficiency.
- It prevents manual intervention and save human efforts.
- It can also be used with advanced programming logic for decision-based tasks.

## Variables

- Variables are used to store values that can be used throughout the shell script.
- Variables makes scripts more flexible and easier to maintain.
- It helps to define value under certain set of words.
- For example: a="hello"(Without Space)  
                  echo \$a

## Loops

- Loops are used to execute a block of commands repeatedly until a specified condition is met.
- Shell supports loops such as for, while and until.
- For example:  
for i in 1 2 3  
do  
    echo \$i  
done

## Functions

- Functions are reusable block of code that performs specific task.
- They help organize scripts and avoid repeating the same code.
- It's easier to read and maintain
- For example:  
Greet(){  
    echo "Hello from Davine Technologies"  
}  
Greet

## Cron Jobs

- Cron Jobs are scheduled task in linux that run commands and scripts automatically at specific time intervals.

- They are commonly used to automate repetitive tasks such as backups, log cleanup, system health checks are configured using the crontab.
- For example:  
`0 8 * * * /home/user/system_info.sh`  
 This runs the script everyday at 8:00 AM.

## **Apache**

- Apache is an open-source web server used to host websites and web applications.
- It processed the HTTP request from the clients and server web pages to web browser.
- Features:
  1. Open-source
  2. Cross-platform
  3. Highly configurable
  4. It supports HTTP and HTTPs

## **Nginx**

- Nginx is a high-performance web server and reverse proxy known for handling a large number of concurrent connections efficiently.
- It is commonly used for load balancing, caching, serving static content.
- Features:
  1. Lightweight
  2. High performance
  3. Supports Reverse proxy
  4. Efficient resource usage

## **Log Management**

- Log management is the process of collecting, storing, monitoring and maintaining log files and generated by the applications and operating system.
- It helps administrator troubleshoot issues, monitor system health, and improves security.
- Common Log file: `/var/log/syslog`, `/var/log/auth.log` .

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